Yuanshuo Du

GitHub YuanshuoDu ▪ LinkedIn yuanshuodu ▪ Email duyuanshuo@gmail.com
Tel +353 89 454 6468 ▪ Location Dublin, Ireland

Skills

ML / Deep Learning: CNN, ResNet50, R-CNN, Deep Learning, TensorFlow, PyTorch, scikit-learn, Python
Programming: Python, Go, Node.js, SQL, R
Data Engineering: PostgreSQL, MongoDB, Apache Spark, Data Visualization
AI Engineering: RAG, Claude API, FastAPI, Prompt Engineering, Docker, GitBook

Dear Hiring Manager,

Combining hands-on computer vision research with applied ML application development, I bring a rare blend of model development and software engineering that bridges the gap between prototype and production. My MSc dissertation applied CNNs (ResNet50, R-CNN) to satellite imagery for real-time parking detection – taking a deep learning model from architecture design through to a containerized microservice. I am eager to bring this end-to-end ML engineering mindset to your team.

From CNN architecture to applied ML systems

At the core of my dissertation, I designed and trained ResNet50 and R-CNN models on satellite and aerial imagery to classify parking space occupancy in real time. I used transfer learning from ResNet50 pretrained weights and achieved strong detection accuracy on the test dataset. I then containerized the inference pipeline with Docker and deployed it on Kubernetes, giving me direct experience with the full lifecycle of computer vision applications – from model architecture through to a serving system.

Building RAG pipelines with large language models

Beyond computer vision, I have built production RAG (Retrieval-Augmented Generation) systems using the Claude API and FastAPI. I implemented the full pipeline: document chunking, embedding generation, semantic search with pgvector in PostgreSQL, and LLM-powered answer synthesis. This gave me direct experience with the full lifecycle of LLM-based applications – from prompt engineering and retrieval tuning to API integration and containerized deployment.

Spatial data analysis and visualization

My GIS project for public amenity analysis in urban areas – published at EDULEARN25 – demonstrates my ability to process large spatial datasets, perform accessibility analysis, and produce publication-quality visualizations. I used Python and R for data processing, and GIS tools for spatial analytics. This work reflects my broader interest in applying data science to real-world infrastructure and public service problems.

Why this role, and why me

I am drawn to roles that sit at the intersection of model research and ML application engineering – where building something that works in a notebook is only the beginning. I bring fluency in Chinese and English (professional), a strong foundation in Python and ML tooling (PyTorch, scikit-learn), and a commitment to ensuring model robustness and reliability in real-world applications. My IT consulting background also means I am comfortable working across engineering, product, and operations teams in Agile environments.

I would welcome the opportunity to discuss how my background in computer vision, RAG systems, and ML application development aligns with your team's needs. Thank you for your time and consideration.

Best regards,

Yuanshuo Du